Ali Vaziri

Machine Learning, Simulation, Control, and Estimation for Physical AI
Ph.D. Student, Mechanical Engineering
Michigan State University

+1 (785) 764–7916
vazirial@msu.edu
GitHub | LinkedIn
Google Scholar

I. Professional Summary

AI/ML engineer and Ph.D. researcher building learning, planning, and control systems for autonomous vehicles, robotics, and physical AI more broadly. My work spans multimodal trajectory prediction, large-scale simulation, physics-guided learning, Bayesian inference, and model predictive control, backed by distributed PyTorch training at scale. At General Motors' Cruise team, I built large-scale data, training, and evaluation pipelines for autonomous-driving simulation. At Mitsubishi Electric Research Labs, I developed physics-informed neural models that combine continuous-time dynamics with physical constraints to deliver accurate predictions from limited real-world data. My graduate research has produced control and estimation methods that run up to 200× faster than standard benchmarks, published in *IEEE Transactions on Robotics*, *Applied Thermal Engineering*, IEEE CDC, and ACC (see Selected Publications, below).

II. Industry Experience

Summer 2026 — Sunnyvale, CA
AI/ML Software Engineering Intern, General Motors – Cruise Team
Scalable ML-Based Motion Reconstruction for Autonomous-Vehicle Simulation

- Built an **end-to-end multimodal ML system** that reconstructs missing vehicles' motion in recorded-driving simulations, enabling more complete and realistic testing of autonomous-driving behavior.
- Built cloud data and featurization pipelines for more than **38,000 highway scenes** (approx. 1 TB of raw tracking data) using Google Cloud Storage, Dataflow, and memory-mapped feature stores.
- Scaled training across 8 H100 GPUs with PyTorch DistributedDataParallel, NCCL, and mixed-precision computation, cutting wall-clock training time **from 12 hours to 3 hours**.
- On a held-out evaluation of approximately 58,000 vehicle trajectories, reduced best-candidate average position error by **24%** relative to the team's prior baseline.
- Reworked full-test evaluation around streaming data, parallel loading, and memory-mapped features, cutting runtime **from 60+ minutes to about 9 minutes** and enabling faster model comparisons and data-quality studies.
- Built a scenario-level collision-detection metric that checks oriented vehicle footprints at 10 Hz, removes duplicate collision pairs, and complements trajectory accuracy with collision, smoothness, and timing measurements.

Summer 2024 — Cambridge, MA
ML and Optimization Research Intern, Mitsubishi Electric Research Laboratories (MERL)
Machine Learning Modeling and Optimization

- Developed a physics-constrained Neural ODE–GRU model for vapor-compression HVAC systems, improving forecasting accuracy by **14%** and runtime by **5.7**× relative to the prior modeling approach.
- Combined continuous-time neural dynamics, recurrent sequence modeling, and known physical constraints to obtain accurate ML predictions from limited data.
- Delivered reusable PyTorch software covering the full pipeline: data processing, model training, evaluation, and GPU-accelerated control experiments.

III. Education

Ph.D. — Mechanical Engineering – University of Kansas 2023–2025
Transferred to Michigan State University 2026–present
Research focus: Optimal inferential control of complex, large-scale systems

M.S. — Mechanical Engineering – University of Kansas 2023–2025
Earned en route to Ph.D. – GPA: 4.00/4.00

Thesis: *Heavy-Tailed Bayesian Motion Planning for Autonomous Vehicles*

B.S. Marine Engineering – Sharif University of Technology 2016–2021
GPA: 3.55/4.00 – Thesis: *Collision-Free Marine Waste-Collection Robot*

IV. Academic Research Experience

2023–present **Graduate Research Assistant** — University of Kansas (2023–2025); Michigan State University (2026–present)

Scientific Machine Learning and Intelligent Control/Estimation

- Developed a new inferential framework for efficient, fast optimal-control methods such as model predictive control (MPC), applied to neural-network-based vehicle models, high-dimensional fluids, and thermofluid systems; tested configurations ran up to **200× faster** than CasADi/IPOPT and MPPI benchmarks.
- Developed continuous-time optimal control of Neural ODEs for soft-robot manipulators.
- Developed GPU-compatible tensor-variate ensemble Kalman smoothing for high-dimensional spatial models, achieving up to **18.6× faster computation** and roughly 11× lower memory use than a vectorized GPU baseline.
- Designed robust motion planning with Student's-*t* distributions, sequential Monte Carlo, and ensemble Kalman smoothing to handle non-Gaussian uncertainty and outliers.
- Trained GRU and ResNet vehicle-dynamics models on real-world driving data and connected them to Bayesian planning and conventional trajectory optimization pipelines.

V. Selected Publications

- I. Askari, **A. Vaziri**, X. Tu, S. Zeng, and H. Fang, "Model Predictive Inferential Control of Neural State-Space Models for Autonomous Vehicle Motion Planning," *IEEE Transactions on Robotics*, vol. 41, pp. 3202–3222, 2025. [Paper]
- **A. Vaziri** et al., "Modular Deep Learning Framework for Vapor-Compression HVAC Systems: Scalable, Efficient, and Physically Consistent Modeling," *Applied Thermal Engineering*, 2026. [Paper]
- **A. Vaziri** et al., "Continuous-Time Optimal Control of Neural ODEs via Bayesian Inference," *IEEE CDC*, 2026.
- **A. Vaziri**, et al., "Bayesian Inferential Motion Planning Using Heavy-Tailed Distributions," *ACC*, 2025. [Paper]
- **A. Vaziri** et al., "Optimal Inferential Control of Convolutional Neural Networks," *ACC*, 2025. [Paper]

VI. Technical Skills

Machine Learning and Modeling: Deep learning, generative modeling, multimodal learning, sequence and time-series modeling, convolutional neural networks (CNNs), recurrent neural networks, Neural ODEs, probabilistic modeling, physics-guided learning, scientific machine learning, system identification

Control, Planning, and Estimation: Model predictive control, Bayesian inference, trajectory optimization, robust motion planning, Kalman filtering and smoothing (EKF, UKF, EnKF/EnKS), particle filtering, sequential Monte Carlo, uncertainty-aware planning

Programming and ML Systems: Python, C/C++, CUDA, PyTorch, TensorFlow, NumPy, SciPy, DistributedDataParallel (DDP), NCCL, mixed-precision training, distributed GPU computing, performance profiling, unit and integration testing

Simulation and Physical AI: Road-to-Sim, Webviz, scenario conversion, world/scene reconstruction, simulation testing, trajectory evaluation, collision and quality metrics

Robotics and Optimization Tools: ROS, MuJoCo, PyBullet, CasADi, IPOPT, MATLAB/Simulink, SolidWorks

Cloud, Data, and Development Tools: Google Cloud Storage, Dataflow, RoboFlow, memory-mapped datasets, Bazel, Linux, Git